SNAP — Research Analysis and Product Research Framework

1. Research Overview

SNAP is proposed as an AI-powered research assistant designed to help users explore, understand, and synthesize information across different levels of complexity. The system aims to address a common limitation of existing AI and search-based research tools: users often receive information, but are required to manually determine the appropriate depth, structure, sources, and next steps for their research.

The research focuses on understanding how an AI research assistant can transform an open-ended research question into a structured and progressively deeper research workflow.

The primary objective is not simply to generate answers, but to investigate how structured interaction, How a answer could be given with a ease and is easily understandable
For making that wase it has 4 types of researching mode for specific task
> Explain
> Compare
> synthesize
> Deep Drive

2. Problem Analysis

2.1 Existing Research Workflow

A typical research process requires users to:

1. Define the research question.
2. Search for relevant information.
3. Open and evaluate multiple sources.
4. Compare information from different sources.
5. Understand unfamiliar concepts.
6. Organize the collected information.
7. Identify unanswered questions.
8. Conduct additional searches.
9. Synthesize the findings into a usable conclusion.

This process can become inefficient when users are unfamiliar with the topic or when the research requires multiple levels of explanation.

AI tools have reduced some of this friction, but they introduce another challenge: users may receive an answer without a clearly defined research structure, depth level, or exploration path.

2.2 Identified Research Gap

The existing ecosystem contains several categories of tools, but each generally emphasizes a particular aspect of research:

> Search engines prioritize information discovery.
> General-purpose AI assistants prioritize conversation and answer generation.
> Academic research platforms prioritize scholarly literature.
> AI search engines prioritize rapid information retrieval and synthesis.

This creates an opportunity for a system that combines these capabilities into a structured, user-controlled research workflow.

SNAP therefore focuses on the following gap:

> How can an AI system adapt the depth, structure, and direction of research assistance according to the user's knowledge level and research objective

3. Target Users

SNAP is designed around four primary user groups.

3.1 Undergraduate and Postgraduate Students

Students often need to understand unfamiliar subjects before progressing toward technical or academic material.

Primary requirements:

> Simple explanations
> Topic overviews
> Step-by-step exploration
> Suggested follow-up questions
> Reliable sources
> Progressive increase in technical depth

3.2 Researchers

Researchers require deeper information and efficient synthesis of complex topics.

Primary requirements:

> Technical analysis
> Literature discovery
> Comparison of approaches
> Research gaps
> Source attribution
> Context-aware follow-up research

3.3 Professionals

Professionals may need to quickly understand an unfamiliar technology, market, concept, or domain.

Primary requirements:

> Concise summaries
> Key findings
> Industry context
> Comparisons
> Practical implications
> Rapid exploration

3.4 Curious Learners

Users who are exploring a topic without a predefined academic or professional objective require an accessible way to move from basic concepts to deeper understanding.

Primary requirements:

> Beginner-friendly explanations
> Structured learning paths
> Related topics
> Follow-up questions
> Progressive exploration

4. Core Research Questions

The research should investigate the following questions:

RQ1 — Structured Research Assistance

How can large language models provide research assistance in a consistent and structured format rather than producing only conversational responses?

This investigates whether predefined response structures such as summaries, key findings, comparisons, limitations, sources, and follow-up questions can improve research usability.

RQ2 — Research Depth

How should an AI research assistant adapt the level of explanation to different user requirements?

The research will investigate whether predefined research modes can effectively serve users with different levels of expertise.

RQ3 — Prompt and Output Reliability

Which prompting and output-structuring strategies produce consistent and usable research responses?

This includes evaluating structured JSON outputs, schemas, system instructions, validation mechanisms, and fallback handling.

RQ4 — Conversational Context

How does maintaining context across multiple interactions affect the quality and efficiency of research sessions?

The goal is to determine whether contextual memory allows SNAP to avoid repeated explanations and generate more relevant follow-up research.

RQ5 — Follow-Up Guidance

Can automatically generated follow-up questions improve the depth and continuity of a research session?

This evaluates whether the system can help users move from a broad question toward increasingly specific research questions.

RQ6 — User Interface

Which interface patterns best support AI-assisted research workflows?

Potential interfaces include:

> Research canvas
> Structured answer cards
> Source panels
> Topic maps
> Follow-up question suggestions
> Research history
> Expandable levels of detail

5. Proposed Research Modes

Instead of providing the same response structure for every user, SNAP proposes four research modes.

Mode	Purpose	Expected Output
Explain	Understand a topic	Simple explanation, examples, key concepts
Explore	Discover a topic	Overview, related concepts, questions, sources
Analyze	Understand a topic deeply	Technical analysis, comparisons, evidence, limitations
Research	Conduct structured investigation	Research question, findings, sources, gaps, follow-up directions

The modes are intended to provide progressive control over research depth , rather than forcing users to manually construct increasingly detailed prompts.

6. Proposed SNAP Research Workflow

The research assistant can follow a structured pipeline:

User Question → Intent Detection → Research Mode → Information Retrieval → Source Evaluation → Information Synthesis → Structured Response → Follow-Up Suggestions → Contextual Continuation**

Step 1 Question Understanding

The system identifies the user's topic, objective, expected depth, and potentially missing context.

Step 2 Mode Selection

The system either uses the user's selected mode or recommends an appropriate mode based on the research objective.

Step 3 Information Retrieval

Relevant information is collected from appropriate sources.

Step 4 — Source Evaluation

Sources should be evaluated based on factors such as relevance, credibility, recency, and relationship to the research question.

Step 5 — Synthesis

The retrieved information is transformed into a structured response rather than simply presented as a collection of search results.

Step 6 — Structured Presentation

The response may include:

> Executive summary
> Key findings
> Detailed explanation
> Evidence/sources
> Comparisons
> Limitations
> Open questions
> Suggested next steps

Step 7 — Follow-Up Research

SNAP generates potential next questions based on the user's current research context.

Step 8 — Session Continuity

Previous questions, selected topics, and research context are retained to support multi-turn research.

7. Competitive Landscape

Existing tools can be grouped according to their primary research approach.

Tool Category	Primary Strength	Limitation / Opportunity for SNAP
General AI Assistants	Flexible conversation and broad knowledge	Users may need to manually structure research workflows
AI Search Engines	Search + generated synthesis	Research depth and workflow control may be limited
Academic Research Tools	Scholarly literature discovery	Primarily designed for academic research
Search Engines	Broad information discovery	Requires significant manual reading and synthesis
Knowledge Platforms	Accessible topic overviews	Limited adaptive research depth and conversational exploration

Competitive Positioning

SNAP should not attempt to compete solely on the ability to generate answers. Instead, its differentiation should focus on the “research workflow”

The proposed differentiators are:

1. Multiple research modes
2. Controllable research depth
3. Structured research outputs
4. Context-aware follow-up questions
5. Persistent research sessions
6. Source-aware synthesis
7. Research canvas for organizing findings

8. Key Research Hypotheses

The current project should treat these as hypotheses to be tested rather than established facts.

H1 — Structured Output

Users will be able to understand and reuse research findings more effectively when information is presented in a consistent structure rather than as an unstructured conversational response.

H2 — Research Modes

Providing explicit research modes will reduce the amount of prompt engineering required from users and improve control over response depth.

H3 — Follow-Up Questions

Contextually relevant follow-up suggestions will increase the number of meaningful research iterations within a session.

H4 — Context Preservation

Maintaining research context across multiple interactions will reduce repetitive queries and improve the relevance of subsequent responses.

H5 — Source Attribution

Clearly presented source information will improve users' ability to evaluate and verify AI-generated research.

9. Evaluation Framework

To make the research experimentally workable, SNAP should be evaluated using measurable criteria rather than subjective claims alone.

9.1 Response Quality

Evaluate responses using:
> Accuracy
> Relevance
> Completeness
> Clarity
> Structure
> Source quality

Each criterion can be scored using a predefined rating scale.

9.2 Research Efficiency

Measure:

> Time required to reach a useful conclusion
> Number of searches/questions required
> Number of sources reviewed
> Number of repeated queries
> Time spent organizing information

9.3 User Experience

Evaluate:

> Ease of use
> Perceived control
> Understanding of research depth
> Satisfaction with response structure
> Usefulness of follow-up suggestions

9.4 Multi-Turn Performance

Compare single-turn and multi-turn research sessions to determine whether context preservation improves:

> Relevance
> Continuity
> Question specificity
> Research depth
> Reduction in repeated information

10. Proposed Experimental Method

A small user study can be conducted with students and other target users.

Experimental Groups

Group A — Conventional AI Interaction

Users receive a general-purpose conversational interface without predefined research modes.

Group B — SNAP Structured Research

Users receive the same type of research capability but with:

> Research modes
> Structured outputs
> Follow-up questions
> Persistent context
> Research canvas

Example Task

Participants could be given a research topic such as:

> "How are large language models being used in robotics?"

Participants would be asked to:

1. Understand the basic concept.
2. Identify major applications.
3. Compare approaches.
4. Identify limitations.
5. Find potential future research directions.

The performance of both approaches can then be compared.

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11. Success Metrics

The initial prototype can define the following measurable goals:

Metric	Measurement
Response relevance	User/evaluator rating
Response structure	Schema validation + evaluator rating
Research completion time	Minutes per task
Follow-up usefulness	Rating of suggested questions
Context relevance	Multi-turn evaluator score
Source usefulness	Source quality/relevance rating
User satisfaction	Post-task questionnaire
Output reliability	Percentage of responses passing required structure

These metrics can later be converted into a formal evaluation dataset and benchmark.

12. Expected Research Contribution

The SNAP project aims to investigate a broader question:

> How can AI systems move from being answer-generation tools toward structured research assistants that actively support the research process?

The expected contribution is a prototype and evaluation framework demonstrating how:

> Research depth can be controlled through predefined modes.
> Structured outputs can improve information organization.
> Conversational context can support iterative research.
> Follow-up questions can guide exploration.
> Source attribution can support verification.
> A research canvas can provide continuity across a research session.

The research should therefore evaluate SNAP not simply on whether it can generate good answers, but on whether it can help users conduct research more efficiently, systematically, and transparently.

13. Recommended Research Roadmap

Phase 1 — Literature and Existing Tool Analysis

> Study AI research assistants and AI search systems.
> Analyze existing research workflows.
> Identify common limitations.
> Review research on human-AI interaction and information retrieval.

Phase 2 — System Design

> Define research modes.
> Design response schemas.
> Define context-management strategy.
> Design source attribution system.
> Design research canvas.

Phase 3 — Prototype Development

> Implement research pipeline.
> Implement structured output generation.
> Add validation.
> Implement multi-turn context.
> Implement follow-up question generation.

Phase 4 — Evaluation

> Create standardized research tasks.
> Recruit test users.
> Compare SNAP with conventional AI interaction.
> Measure accuracy, efficiency, usability, and research depth.

Phase 5 — Analysis

> Analyze quantitative results.
> Collect qualitative user feedback.
> Identify system limitations.
> Refine the research workflow

Final Research Direction

The strongest way to position SNAP is not:

Is certainly not that it’s the best AI tools that give best output

Instead, position it as:

> "A structured AI research assistant that adapts research depth, maintains contextual continuity, organizes findings, and guides users through iterative research, you don’t have to hunt for comparisons, deep drive through topic all of it would be done by SNAP

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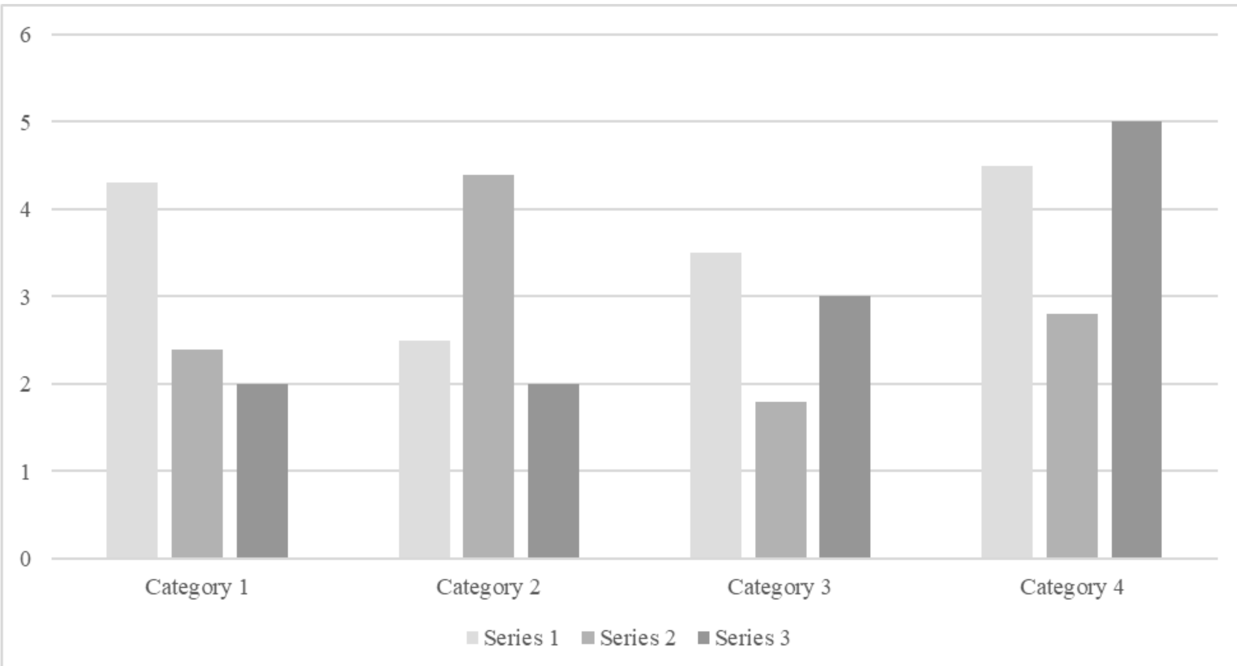

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